1 Subspace Generated by a Set

Suppose we are given a subset W of a vector space V. If W is not a subspace of V, then we can make W into a subspace by adding some vectors to the set W. That is, we would like to extend W to some subspace of V which contains W (we know there is at least one such subspace, namely V itself). The most economical choice would be to add the minimum number of additional vectors to make W into a subspace; that is, to extend W to the smallest subspace which contains it, which, as we have seen, is the intersection of all subspaces of V containing W. This subspace is called the *subspace generated by* W.

1. If V is any vector space, what is the subspace generated by $\emptyset$?

Given an arbitrary subset W of V, the definition above does not make it very easy to find the subspace generated by W. For example, suppose

$$W = \{e^t, \sin(t), \cos(t)\} \subset \mathbb{R}^{\mathbb{R}}.$$

To find the subspace of $\mathbb{R}^{\mathbb{R}}$ generated by W, we need to find all subspaces of $\mathbb{R}^{\mathbb{R}}$ which contain W and then take their intersection. This seems like a very difficult task. However, we can describe the subspace generated by W much more explicitly. The key is to think about building up this subspace out of W. To extend the set W above to a subspace, first of all we must include the zero vector. A subspace must be closed under scalar multiplication, so we must include all scalar multiples of the vectors in W:

$$\alpha_1 e^t, \alpha_2 \sin(t), \alpha_3 \cos(t), \quad \alpha_i \in \mathbb{R}.$$

A subspace must also be closed under addition, so we must also include all possible sums of such vectors:

$$\alpha_1 e^t + \alpha_2 \sin(t) + \alpha_3 \cos(t), \quad \alpha_i \in \mathbb{R}.$$

A vector of this form is called a *linear combination* of the vectors $e^t, \sin(t), \cos(t)$.

Definition 1.1 (Linear combination)**.** A *linear combination* of a collection $v_1, \ldots, v_n$ of vectors in a vector space V is a vector of the form

$$\alpha_1 v_1 + \cdots + \alpha_n v_n,$$

where each $\alpha_k \in F$.

In our example above, the subspace generated by W should therefore be the set of all linear combinations of W:

$$\{\alpha_1 e^t + \alpha_2 \sin(t) + \alpha_3 \cos(t), \quad \alpha_i \in \mathbb{R}\}$$

as we have added just the vectors we need to make W into a subspace, and nothing else. We will prove below that this is indeed the case.

Definition 1.2 (Span)**.** Given a subset W of a vector space V, the set of all linear combinations of vectors in W is called the *span* of W:

$$\operatorname{span}(W) := \left\{ \sum_{i=1}^{n} \alpha_i w_i : \alpha_i \in F, w_i \in W \right\}.$$

Note that all sums on the right-hand side are finite, by definition of a linear combination. In particular, if $W = \{w_1, \ldots, w_n\}$ is finite, then we write

$$\operatorname{span}(w_1, \ldots, w_n) = \{\alpha_1 w_1 + \cdots + \alpha_n w_n : \alpha_i \in F\}.$$

We also define $\operatorname{span}(\emptyset) = \{0\}$. [1]

In this language, in our example above

$$\operatorname{span}(W) = \{\alpha_1 e^t + \alpha_2 \sin t + \alpha_3 \cos t : \alpha_i \in \mathbb{R}\}.$$

Before proving that $\operatorname{span}(W)$ is the subspace generated by W, let's work out a few more examples.

2. Let $W = \{(2,1,-3),(1,-2,4)\} \subset \mathbb{R}^2$. Determine whether each of the following vectors is in $\operatorname{span}(W)$.
   (a) $(17,-4,2)$
   (b) $(17,-4,5)$

3. If $W = \{t^n\}_{n=0}^{\infty} \subset \mathbb{R}^{\mathbb{R}}$, then what is $\operatorname{span}(W)$?

[1] This will allow us to identify, by Proposition 1.3 below, $\operatorname{span}(W)$ with the subspace generated by W for *all* subsets of W, without having to make an exception for $\emptyset$.

Proposition 1.3. *Let V be a vector space and W a subset of V.*

(a) $span(W)$ is a subspace of V containing W.

(b) If $\{W_\alpha\}_{\alpha\in\mathscr{A}}$ is the collection of subspaces of V containing W, then $span(W) = \bigcap_{\alpha\in\mathscr{A}} W_\alpha$.

(c) If W is a subspace of V, then $span(W) = W$.

3. Prove Proposition 1.3.

4. Let $v = (1,2,-1)$ and $w = (6,4,2)$ in $\mathbb{R}^3$. What are all the vectors $b = (b_1,b_2,b_3)$ in $\text{span}(v,w)$?

Definition 1.4 (Spanning set)**.** If $\text{span}(W) = V$, then we say that W *spans* V. We also say that W *is a spanning set for* V.

5. Show that the set of vectors $(1,0,0),(0,1,0),(0,0,1)$ spans F^3. More generally, the n vectors, e_i, with a 1 in the nth position and 0s elsewhere, span F^n.

6. (a) Show that the set $\{1,t,t^2,\ldots,t^n\}$ spans the vector space $P_n(F)$ of polynomials of degree at most n with coefficients in F.
   (b) Show that the set $\{1,t,t^2,\ldots,t^k,\ldots\}$ spans the vector space $P(F)$ of all polynomials with coefficients in F.

2 Finite-Dimensional Vector Spaces

Definition 2.1 (Finite-dimensional vector spaces)**.** A vector space is called *finite-dimensional* if it has a finite spanning set. Otherwise, it is called *infinite-dimensional*.

Example 2.2 (Examples of finite-dimensional vector spaces)**.**

(a) By definition, $\{0\}$ is finite-dimensional.

(b) We found a finite spanning set for F^n in Problem 5, so F^n is finite-dimensional.

(c) We found a finite spanning set for $P_n(F)$ in Problem 6, so $P_n(F)$ is finite-dimensional.

7. Prove that the vector space $P(F)$ of all polynomials with coefficients in F is infinite-dimensional. (*Hint*: Let W be any finite set of polynomials with coefficients in F. Find a polynomial that is not in span(W).)

7. Consider F^ω, the vector space of all sequences of elements of F. We will prove later that F^ω is infinite-dimensional. For now, generalize the spanning set for F^n that we found in Problem 5 by letting e_i denote the sequence with a 1 in the ith entry and zeros elsewhere. Explain why the set of e_is is not a spanning set for F^ω.